



Crystal Structure and Li-Ion Conductivity of $\text{LiGa}_{1-x}\text{Al}_x\text{GeO}_4$ Phenacite Compounds with $0 \leq x \leq 1$

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Compounds of $\text{LiGa}_{1-x}\text{Al}_x\text{GeO}_4$ with $0 \leq x \leq 1$ were examined using powder X-ray diffraction measurements and electrochemical impedance spectroscopy, to clarify their structures and Li-ion conductivity (σ_{Li}) as a function of x , and to compare with those for a wide variety of AB_2O_4 -type oxides. The $x = 0$ and 0.25 samples were found to exhibit a phenacite structure with $R\bar{3}$ space group, whereas the $x \geq 0.5$ samples were found to exhibit an ordered phenacite structure with $R3$ space group due to the 1:1 cation order between $\text{Ga}^{3+}/\text{Al}^{3+}$ and Ge^{4+} ions. The σ_{Li} value for the $x = 0$ sample was $2.81(1) \times 10^{-5} \text{ S}\cdot\text{cm}^{-1}$ at 973 K. Owing to the 1:1 cation order, both σ_{Li} and activation energy (E_a) were slightly improved at $x = 1$. The structural details of $\text{LiGa}_{1-x}\text{Al}_x\text{GeO}_4$, possible conduction pathway of Li^+ ions, and correlation between crystal structure and σ_{Li} (E_a) are presented.

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The mineral name phenacite has its origins from the Greek word “phenakos” which means “a deceiver”, due to its similarity in color with quartz. Beryllium orthosilicate Be_2SiO_4 , which is the parent compound of phenacite, is reported to crystallize into rhombohedral structure with $R\bar{3}$ space group.¹ As illustrated in Fig. 1, two nonequivalent Be^{2+} ions occupy tetrahedral 18f sites, and Si^{4+} ions occupy the other tetrahedral 18f site. The BeO_4 tetrahedra and SiO_4 tetrahedra connect via edge sharing, forming 4-tetrahedron and 6-tetrahedron rings. Among AB_2O_4 -type oxides, phenacite has 3/8 tetrahedral holes in the closest packing of the cations, while olivine (Mg_2SiO_4) and spinel (MgAl_2O_4) have 1/8 tetrahedral holes and 1/2 octahedral holes, respectively.²

Besides Be_2SiO_4 , there is a wide variety of phenacite compounds containing lithium atoms; for instance, LiGaGeO_4 ,³ LiAlGeO_4 ,^{3–5} Li_2WO_4 ,⁶ Li_2MoO_4 ,^{7,8} and LiZnVO_4 .⁹ Tetravalent chromium-doped LiAlGeO_4 is a promising material for use as an active medium for near-infrared solid-state lasers.⁵ Li_2MoO_4 shows a reversible electrochemical reactivity in nonaqueous lithium cells, leading to an application of a negative electrode material for lithium-ion batteries (LIBs).⁸ Note that the Si site (18f) in Be_2SiO_4 corresponds to the Li site in LiGaGeO_4 ³ and LiAlGeO_4 ,^{3–5} the W site in Li_2WO_4 ,⁶ the Mo site in Li_2MoO_4 ,^{7,8} and the V site in LiZnVO_4 .⁹ Thus, lithium atoms can occupy three different 18f sites in the phenacite structure.

In this paper, we focused on the $\text{LiGa}_{1-x}\text{Al}_x\text{GeO}_4$ (LGAGO) series with $0 \leq x \leq 1$, because their related compounds $\text{Li}_{2.5}\text{Al}_{0.5}\text{SiO}_4$,¹⁰ $\text{Li}_3\text{AlGeO}_5$,¹¹ $\text{Li}_{2.5}\text{Ga}_{0.5}\text{GeO}_4$,¹² and $\text{Li}_{4-3x}\text{Al}_x\text{GeO}_4$ ¹³ have attracted much attention as solid electrolytes for all-solid-state LIBs. Despite extensive studies on the phenacite compounds described above,^{1,3–13} the structural chemistry and nature of Li-ion conduction in the LGAGO series still remain unclear. One of the benefits of LGAGO is the low heating temperature for the synthesis compared to other solid electrolytes. The LGAGO and related compounds are usually prepared below 1000°C ,^{3–13} whereas current solid electrolytes such as $\text{Li}_{1-x}\text{La}_{2/3-x}\text{TiO}_3$ (LLTO)¹⁴ and $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ ¹⁵ are prepared above 1200°C . Since electrode materials for all-solid-state LIBs are typically synthesized below 1000°C ,¹⁶ similar heating temperature enables to sinter all the components in a body, providing a well-connected solid-solid interface. We performed powder X-ray diffraction (XRD) measurements and electrochemical impedance spectroscopy (EIS) for the LGAGO samples with $0 \leq x \leq 1$ to clarify the changes in crystal structure and Li-ion conductivity (σ_{Li}) as a function of x . This information will facilitate the design of advanced solid electrolytes with high σ_{Li} values and further the understanding of the structural chemistry of AB_2O_4 compounds, such as phenacite, olivine, and spinel.

Experimental

Powdered LGAGO samples with $x = 0, 0.25, 0.5, 0.75$, and 1 were synthesized by a solid-state reaction technique. A stoichiometric mixture of Li_2CO_3 (Wako Pure Chemical Industries, Ltd.), Ga_2O_3 (Kojundo Chemical Laboratory Co., Ltd.), $\text{Al}(\text{OH})_3$ (Kojundo Chemical Laboratory Co., Ltd.), and GeO_2 (Kojundo Chemical Laboratory Co., Ltd.) was mixed using a pestle and mortar. Each powder was pressed into a pellet of 23 mm diameter and ~ 5 mm thickness using conventional uniaxial pressing (UP). The pellet was pre-heated at 973 K for 12 h in an oxygen flow, and then crushed into a powder again. To obtain the LGAGO pellets, we employed the same procedure for $\text{Li}[\text{Li}_{1/3}\text{Ti}_{5/3}]\text{O}_4$.¹⁷ The powder was pressed at 200 MPa for 3 min by a cold isostatic pressing (CIP), followed by pressing using UP again. During CIP, the pellet was packed into a nylon bag to avoid unwanted reactions with water. The pellet obtained was finally heated at 1273 K for 12 h in an oxygen flow. Heating and cooling rates were 200 and 60 K/h, respectively.

Crystal structures of the LGAGO samples were investigated by powder XRD measurements (D8 ADVANCE, Bruker AXS, Inc.) equipped with $\text{Cu-K}\alpha$ radiation. Structural parameters, such as lattice parameters and bond lengths, were determined by Rietveld refinements with the RIETAN-FP program.¹⁸ Crystal structure schematics were drawn by VESTA.¹⁹ Differential scanning calorimetry (DSC) analysis was conducted for the $x = 0, 0.5$, and 1 samples to investigate possible structural phase transitions (Thermo plus, Rigaku Co. Ltd.). The particle morphology of LGAGO was observed by scanning electron microscopy (SEM; S-3600 N, Hitachi High-Technologies Co., Ltd.).

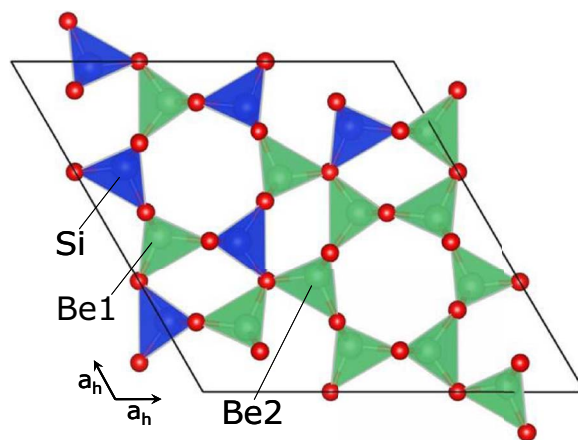


Figure 1. Crystal structure of Be_2SiO_4 from the viewpoint of c_h -axis. Be_2SiO_4 has a phenacite structure with $R\bar{3}$ space group, in which two nonequivalent Be^{2+} ions (Be1 and Be2) occupy tetrahedral 18f sites and Si^{4+} ions occupy the other tetrahedral 18f site.

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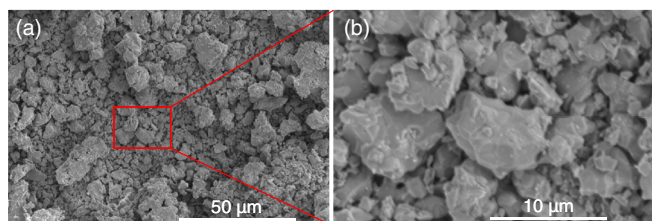


Figure 2. (a) SEM image for the $\text{LiGa}_{1-x}\text{Al}_x\text{GeO}_4$ sample with $x = 0$ at the $50\ \mu\text{m}$ scale and (b) its magnified image at the $10\ \mu\text{m}$ scale.

To determine σ_{Li} for the LGAGO samples, EIS was performed in the temperature range between 673 and 973 K under nitrogen flow at $500\ \text{mL}\cdot\text{min}^{-1}$ (1260 impedance/gain phase analyzer, Solartron). The procedure used for EIS was similar to that in our previous report.¹⁷ Each LGAGO pellet was cut into a cylindrical form of $\sim 10\ \text{mm}$ diameter and $\sim 0.5\ \text{mm}$ thickness, then polished with waterproof abrasive papers. After depositing a thin Au film on the surface of the sample, two Au terminals were attached to the cylindrical electrode with Au paste. To evaporate organic binders in the Au paste, the electrolyte was annealed at 973 K in 20% O_2/N_2 gas flow before the measurements.

Results

Particle and electrolyte characterization.—Figure 2a shows the SEM image for the $x = 0$ sample at the $50\ \mu\text{m}$ scale. The magnified image at the $10\ \mu\text{m}$ scale is shown in Fig. 2b. The primary particles with $\sim 1\ \mu\text{m}$ diameter aggregate together, forming large secondary particles. The sizes of the secondary particles range between 5 and $20\ \mu\text{m}$. No particular crystal habit, such as hexagonal or columnar, is observed in the primary or secondary particles. As seen in Fig. 2b, the surface of the secondary particles appears to be flat and formed from several fused primary particles. Similar surface morphology was reported for Li_2MoO_4 with phenacite structure, which was prepared at 600°C .⁶ As illustrated in Fig. S1 of the Supplementary Material, non-uniform particle shapes for the primary and secondary particles were observed in the $x = 0.25, 0.5, 0.75$, and 1 samples. However, the size of the secondary particles decreased with increasing x . The average size of the secondary particles was $\sim 5\ \mu\text{m}$ for the $x = 1$ sample.

In response to the decrease in particle size with increasing x , the bulk density (d_{bulk}) for the EIS measurements decreased with increasing x . Here, d_{bulk} was calculated by using the weight and dimensions (volume) of each electrolyte. d_{bulk} for the $x = 0, 0.25, 0.5, 0.75$, and 1 samples was 2.69(1), 2.31(1), 2.12(1), 2.01(1), and 1.81(1) $\text{g}\cdot\text{cm}^{-3}$, respectively. Comparison of these data with those determined by XRD measurements (d_{XRD}) is discussed in the next section.

XRD measurements.—Figure 3 shows the results of the Rietveld refinements for the LGAGO samples with (a) $x = 0$, (b) $x = 0.5$, and (c) $x = 1$. Those for the $x = 0.25$ and 0.75 samples are shown in Fig. S2 of the Supplementary Material. The most intense diffraction line is the 140 diffraction for the $x \leq 0.75$ samples, but it is the 220 diffraction line for the $x = 1$ sample. The full width at half maximum (FWHM) of the 140 diffraction line is 0.0883, 0.1226, 0.1760, 0.1445, and 0.1011° for $x = 0, 0.25, 0.5, 0.75$, and 1, respectively. The XRD pattern for the $x = 0$ sample is assigned as the phenacite structure as reported previously;³ i.e., Li^+ ions, two nonequivalent Ga^{3+} ions (Ga1 and Ga2), two nonequivalent Ge^{4+} ions (Ge1 and Ge2), and four nonequivalent O^{2-} ions (O1 , O2 , O3 , and O4) occupy the tetrahedral $18f$ sites in $R\bar{3}$ space group (see Table I). The $x = 0$ sample is not in a single phase of the phenacite structure, but contains LiGa_5O_8 and Li_2GeO_3 impurities. Here, LiGa_5O_8 crystallizes into an ordered spinel structure with $P4_332$ (or $P4_132$) space group,²⁰ while Li_2GeO_3 has an orthorhombic structure with $Cmc2_1$ space group.²¹ The weight fraction (W) of these impurities is calculated as 2.9(1)% for LiGa_5O_8 and 1.7(1)% for Li_2GeO_3 . The reliability index of the R -weighted pat-

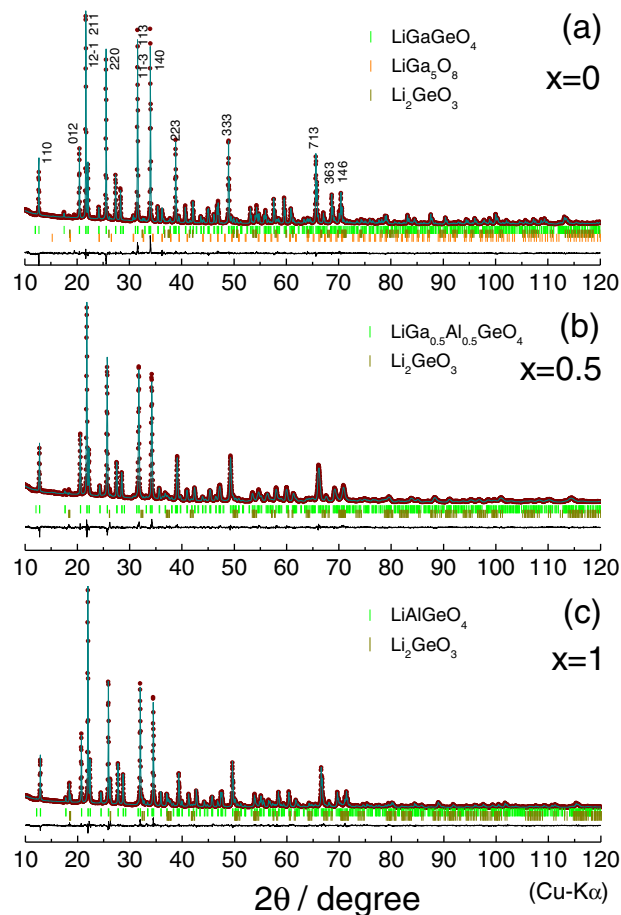


Figure 3. Results of the Rietveld refinements for the $\text{LiGa}_{1-x}\text{Al}_x\text{GeO}_4$ samples with (a) $x = 0$, (b) $x = 0.5$, and (c) $x = 1$. The Rietveld refinements were performed using $R\bar{3}$ space group for the $x = 0$ sample, but using $R3$ space group for the $x = 0.5$ and 1 samples.

tern (R_{wp}) and the goodness-of-fit parameter (S) for the refinement are 5.93% and 1.77, respectively. The lattice parameters in the hexagonal setting are determined to be $a_h = 13.941(1)\ \text{\AA}$ and $c_h = 9.302(1)\ \text{\AA}$, being consistent with the previous report on LiGaGeO_4 .³

The situation for the $x = 0.25$ sample is similar with that for the $x = 0$ sample except for the occupation of the $18f$ site. The XRD pattern of $x = 0.25$ is identified as the phenacite structure with $R\bar{3}$ space group, in which Ga^{3+} , Al^{3+} , and Ge^{4+} ions occupy two different tetrahedral $18f$ sites with the occupancy ratio of $\text{Ga}^{3+}/\text{Al}^{3+}/\text{Ge}^{4+} = 0.375/0.125/0.5$ (see Fig. S2 and Table S1 of the Supplementary Material). The W values for LiGa_5O_8 and Li_2GeO_3 were determined as 3.9(1) and 1.2(1)%, respectively.

Before going to the results for the $x = 0.5$ and 0.75 samples, we describe the result for the $x = 1$ sample. In the case of $x = 1$, the Rietveld analysis using $R\bar{3}$ space group provided relatively poor R -indices ($R_{\text{wp}} = 10.26\%$ and $S = 3.10$). These R -indices were not improved even if we employed possible impurities in $x = 1$, such as $\alpha\text{-LiAlO}_2$,²² $\beta\text{-LiAlO}_2$,²³ or $\gamma\text{-LiAlO}_2$.²⁴ However, looking at the XRD patterns shown in Fig. 3, the ratio of the intensity of the 211 diffraction line to the 220, 113, or 140 diffraction line decreases with increasing x . This is characteristic of an ordered phenacite structure for LiGaSiO_4 , in which Ga^{3+} and Si^{4+} ions arrange along the c_h -axis.³ Because of this cation order, the crystal structure of LiGaSiO_4 lacks an inversion center (i.e., it is non-centrosymmetric), resulting in the lowering of the space group to $R3$. As expected, the XRD pattern for the $x = 1$ sample is well assigned as the ordered phenacite structure with $R3$ space group, with $R_{\text{wp}} = 6.09\%$ and $S = 1.85$ (Fig. 3c). In the $R3$ space group, Li^+ , Al^{3+} , and Ge^{4+} ions occupy two nonequivalent

Table I. Structural parameters determined by Rietveld refinements for the $\text{LiGa}_{1-x}\text{Al}_x\text{GeO}_4$ samples with $x = 0, 0.5$, and 1.

Sample	Atom	Wyckoff position	<i>g</i>	<i>x</i>	<i>y</i>	<i>z</i>	<i>B</i> _{iso}
<i>x</i> = 0	Li	18 <i>f</i>	1	0.451(1)	0.145(1)	0.087(1)	0.54(1)
	Ga1	18 <i>f</i>	0.5	0.121(1)	0.472(1)	0.083(1)	0.41(1)
	Ge1	18 <i>f</i>	0.5	0.121(1)	0.472(1)	0.083(1)	0.41(1)
	Ga2	18 <i>f</i>	0.5	0.020(1)	0.215(1)	0.082(1)	0.55(1)
	Ge2	18 <i>f</i>	0.5	0.020(1)	0.215(1)	0.082(1)	0.55(1)
	O1	18 <i>f</i>	1	0.546(1)	0.409(1)	0.089(1)	0.89(1)
	O2	18 <i>f</i>	1	0.002(1)	0.332(1)	0.072(1)	0.89(1)
	O3	18 <i>f</i>	1	0.211(1)	0.117(1)	0.081(1)	0.89(1)
	O4	18 <i>f</i>	1	0.243(1)	0.459(1)	0.102(1)	0.89(1)
SG: $R\bar{3}$, $a_h = 13.941(1)$ Å, $c_h = 9.302(1)$ Å, $R_{wp} = 5.90\%$, and $S = 1.77$.							
<i>x</i> = 0.5	Li1	9 <i>b</i>	1	0.115(1)	0.480(1)	0.099(1)	0.9(1)
	Li2	9 <i>b</i>	1	0.973(1)	0.782(1)	0.929(1)	0.9(1)
	Al1	9 <i>b</i>	0.5	0.451(1)	0.141(1)	0.096(1)	0.4(1)
	Ga1	9 <i>b</i>	0.5	0.451(1)	0.141(1)	0.096(1)	0.4(1)
	Al2	9 <i>b</i>	0.5	0.539(1)	0.857(1)	0.929(1)	0.4(1)
	Ga2	9 <i>b</i>	0.5	0.539(1)	0.857(1)	0.929(1)	0.4(1)
	Ge1	9 <i>b</i>	1	0.882(1)	0.529(1)	0.931(1)	0.6(1)
	Ge2	9 <i>b</i>	1	0.017(1)	0.215(1)	0.095(1)	0.6(1)
	O1	9 <i>b</i>	1	0.559(1)	0.418(1)	0.087(1)	0.5(1)
	O2	9 <i>b</i>	1	0.459(1)	0.574(1)	0.919(1)	0.5(1)
	O3	9 <i>b</i>	1	0.012(1)	0.339(1)	0.098(1)	0.5(1)
	O4	9 <i>b</i>	1	0.982(1)	0.650(1)	0.935(1)	0.5(1)
	O5	9 <i>b</i>	1	0.195(1)	0.111(1)	0.109(1)	0.5(1)
	O6	9 <i>b</i>	1	0.785(1)	0.867(1)	0.933(1)	0.5(1)
	O7	9 <i>b</i>	1	0.270(1)	0.491(1)	0.100(1)	0.5(1)
	O8	9 <i>b</i>	1	0.770(1)	0.557(1)	0.940(1)	0.5(1)
SG: $R\bar{3}$, $a_h = 13.867(1)$ Å, $c_h = 9.257(1)$ Å, $R_{wp} = 5.67\%$, and $S = 1.69$.							
<i>x</i> = 1	Li1	9 <i>b</i>	1	0.114(1)	0.481(1)	0.099(1)	0.9(1)
	Li2	9 <i>b</i>	1	0.973(1)	0.786(1)	0.928(1)	0.9(1)
	Al1	9 <i>b</i>	1	0.454(1)	0.141(1)	0.088(1)	0.7(1)
	Al2	9 <i>b</i>	1	0.544(1)	0.863(1)	0.921(1)	0.7(1)
	Ge1	9 <i>b</i>	1	0.880(1)	0.528(1)	0.922(1)	0.6(1)
	Ge2	9 <i>b</i>	1	0.017(1)	0.213(1)	0.089(1)	0.6(1)
	O1	9 <i>b</i>	1	0.554(1)	0.411(1)	0.088(1)	0.8(1)
	O2	9 <i>b</i>	1	0.460(1)	0.574(1)	0.915(1)	0.8(1)
	O3	9 <i>b</i>	1	0.008(1)	0.340(1)	0.097(1)	0.8(1)
<i>x</i> = 0.75	O4	9 <i>b</i>	1	0.982(1)	0.647(1)	0.935(1)	0.8(1)
	O5	9 <i>b</i>	1	0.196(1)	0.105(1)	0.110(1)	0.8(1)
	O6	9 <i>b</i>	1	0.785(1)	0.872(1)	0.938(1)	0.8(1)
	O7	9 <i>b</i>	1	0.270(1)	0.482(1)	0.095(1)	0.8(1)
	O8	9 <i>b</i>	1	0.769(1)	0.562(1)	0.943(1)	0.8(1)
SG: $R\bar{3}$, $a_h = 13.772(1)$ Å, $c_h = 9.195(1)$ Å, $R_{wp} = 6.09\%$, and $S = 1.85$.							

9*b* sites, respectively, while O^{2-} ions occupy eight nonequivalent 9*b* sites (Table I). To the best of our knowledge, the cation order between Al^{3+} and Ge^{4+} ions in LiAlGeO_4 has not been reported previously for phenacite compounds.^{1,3,5–13} Furthermore, this cation order is surprising because the ratio of ionic radius (*r*) of M^{3+} to M^{4+} in LiAlGeO_4 differs significantly from the ratio in LiGaSiO_4 . That is, $r_{M^{3+}}/r_{M^{4+}} = 1.00$ for LiAlGeO_4 , whereas $r_{M^{3+}}/r_{M^{4+}} = 1.81$ for LiGaSiO_4 , using $r_{\text{Al}^{3+}} = 0.39$ Å, $r_{\text{Ge}^{4+}} = 0.39$ Å, $r_{\text{Ga}^{3+}} = 0.47$ Å, and $r_{\text{Si}^{4+}} = 0.26$ Å when the coordination number (C. N.) of each *M* ion is 4.²⁴

The XRD pattern for the $x = 0.5$ sample is successfully assigned as the ordered phenacite structure with $R\bar{3}$ space group, as in the case for $x = 1$ (Fig. 3b). As presented in Table I, Ga^{3+} and Al^{3+} ions occupy two nonequivalent 9*b* sites with the occupancy ratio by $\text{Ga}^{3+}/\text{Al}^{3+} = 0.5/0.5$. This arrangement forms the 1:1 cation order between the $\text{Ga}^{3+}/\text{Al}^{3+}$ and Ge^{4+} ions. The XRD pattern for the $x = 0.75$ sample is also identified as the ordered phenacite structure with $R\bar{3}$ space group, as shown in Fig. S2 of the Supplementary Material, with the occupancy ratio by $\text{Ga}^{3+}/\text{Al}^{3+} = 0.25/0.75$ at the 9*b* site. Its structural parameters are listed in Table S1 of the Supplementary Material.

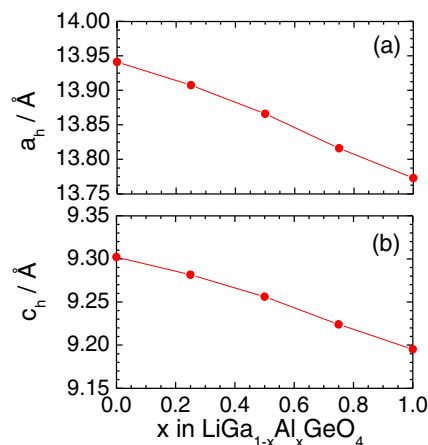
**Figure 4.** Lattice parameters of the a_h - and c_h -axes as a function of x in $\text{LiGa}_{1-x}\text{Al}_x\text{GeO}_4$.

Figure 4 shows the change in a_h - and c_h -length as a function of x in LGAGO. Note that the unit cell in the $R\bar{3}$ space group is identical to that in the $R3$ space group. As x increases from 0, both a_h and c_h decrease almost linearly to $a_h = 13.772(1)$ Å and $c_h = 9.195(1)$ Å at $x = 1$. This corresponds to the linear decrease in r with x ; $r_{\text{Ga}^{3+}} = 0.47$ Å and $r_{\text{Al}^{3+}} = 0.39$ Å.²⁴ From these a_h and c_h values, d_{XRD} is calculated as 4.065(1), 3.894(1), 3.721(1), 3.554(1), and 3.376(1) g·cm⁻³ for the $x = 0, 0.25, 0.5, 0.75$, and 1 samples, respectively. Thus, $d_{\text{bulk}}/d_{\text{XRD}}$ is determined as 0.66(1), 0.59(1), 0.57(1), 0.57(1), and 0.54(1) for the $x = 0, 0.25, 0.5, 0.75$, and 1 samples, respectively. The decrease in $d_{\text{bulk}}/d_{\text{XRD}}$ with x is probably due to the decrease in the average size of the secondary particles, as described in the previous section.

EIS measurements.— σ_{Li} for the LGAGO samples was obtained by fitting their EIS spectra with an equivalent circuit model. Figure 5 shows the Cole-Cole plots for the $x = 0, 0.5$, and 1 samples at (a) 873 K and (c) 673 K, as an example. At 873 K, two deformed semi arcs were observed in the frequency region below and above 1 kHz, while, at 673 K, only one deformed semi arc was observed in the high frequency region above 1 kHz. Thus, the Cole-Cole plot at 873 K was fitted by using a parasitic inductance from the electrical wire and equipment (L_0), a total resistance (R_{total}), its capacitance (C_{total}), an electrode-electrolyte interface resistance (R_i), and its constant phase element (CPE_i), where $Z_{\text{CPE},i}$ is represented as $1/(i\omega Q_i)^{\alpha_i}$; when $\alpha_i = 1$, Q_i has the dimension of capacitance (F), and when $\alpha_i < 1$, the system shows surface heterogeneity.²⁵ The R_{total} (Q_{total}) value, which is obtained by the deformed semi arc in the high frequency above 1 kHz, includes the contributions from a bulk resistance (capacitance) and a grain boundary resistance (capacitance). The equivalent circuit model for the fitting is shown in the inset of Fig. 5a. In contrast, at 673 K, the Cole-Cole plot was fitted only by L_0 , R_{total} , and $\text{CPE}_{\text{total}}$, because the electrode-electrolyte interface element was not observed in the low frequency region below 1 kHz (see the equivalent circuit model in the inset in Fig. 5b). The Cole-Cole plots at 973 and 773 K, which are shown in Fig. S3 of the Supplementary Material, were fitted by the equivalent circuit model at 673 and 873 K, respectively. Hence, using the R_{total} value, σ_{Li} is given by

$$\sigma_{\text{Li}} = \frac{l}{R_{\text{total}} A} \quad [1]$$

where l is the length of the electrolyte and A is the surface area of the electrode. Fitting results of all the parameters are summarized in Tables S2(a) and S2(b) of the Supplementary Material. Figure 6 shows the $\log \sigma_{\text{Li}}$ vs. $1/T$ curve for the LGAGO samples with $0 \leq x \leq 1$. Moreover, its $\ln(\sigma_{\text{Li}} \cdot T)$ vs. $1/T$ curve is shown in Fig. S4 of the Supplementary Information. For all the LGAGO samples, σ_{Li} increases almost exponentially with increasing T . For the $x = 0$ sample,

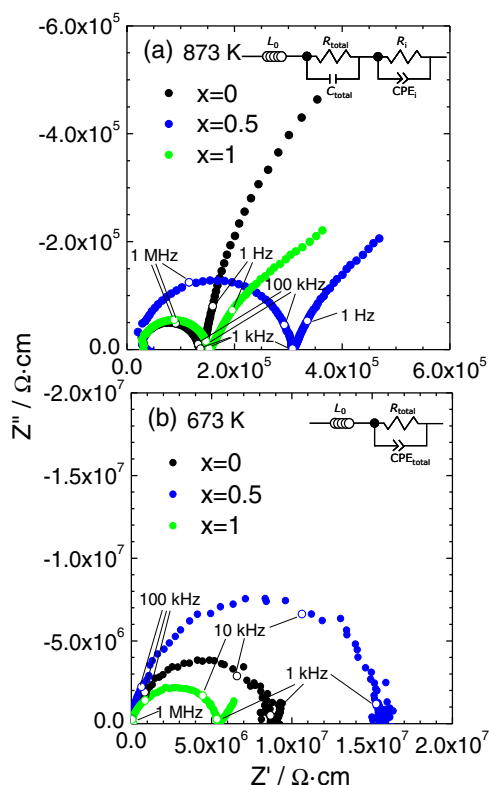


Figure 5. Cole-Cole plots for the $\text{LiGa}_{1-x}\text{Al}_x\text{GeO}_4$ samples with $x = 0, 0.5$, and 1 at (a) 873 K and (b) 673 K . Equivalent circuit model for each Cole-Cole plot is shown in the inset.

for instance, $\sigma_{\text{Li}} = 1.15(1) \times 10^{-7} \text{ S}\cdot\text{cm}^{-1}$ at 673 K and $\sigma_{\text{Li}} = 2.81(1) \times 10^{-5} \text{ S}\cdot\text{cm}^{-1}$ at 973 K . The linear relationship between $\ln(\sigma_{\text{Li}} \cdot T)$ and $1/T$ indicates that no structural phase transition or change in the conduction pathway occurs over the entire T range measured. Indeed, DSC analysis on $x = 0$ clarified no exothermic or endothermic peak up to 1300 K (Fig. S5 of the Supplementary Material). Since Ga^{3+} , Ge^{4+} , and O^{2-} ions in $x = 0$ are rigidly bounded in the framework of phenacite structure, it is most likely that electrical conduction in the $x = 0$ sample is mediated by Li^+ ions. σ_{Li} for $0.25 \leq x \leq 0.75$ is slightly lower than that for $x = 0$, but σ_{Li} for $x = 1$ is close to that for $x = 0$. Since $d_{\text{bulk}}/d_{\text{XRD}}$ decreases with increasing x , a simple comparison of σ_{Li} between the samples is difficult. Intrinsic σ_{Li} ($\sigma_{\text{Li}}^{\text{int}}$) which excludes

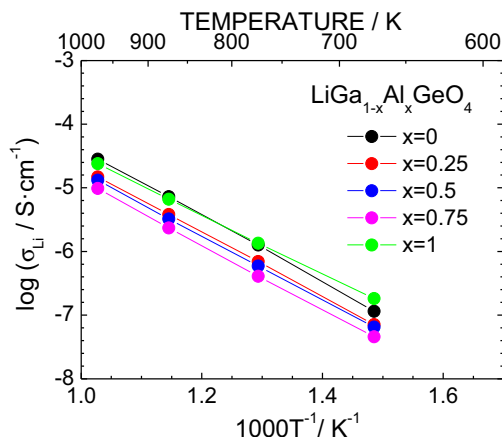


Figure 6. Temperature dependence of Li-ion conductivity (σ_{Li}) for the $\text{LiGa}_{1-x}\text{Al}_x\text{GeO}_4$ samples with $0 \leq x \leq 1$.

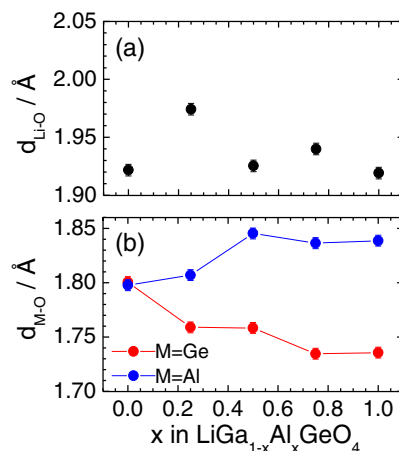


Figure 7. Average bond length between Li^+ and O^{2-} ions in the LiO_4 tetrahedron ($d_{\text{Li-O}}$) and (b) average bond length between M and O^{2-} ions in the MO_4 tetrahedron ($d_{\text{M-O}}$) as a function of x in $\text{LiGa}_{1-x}\text{Al}_x\text{GeO}_4$. M1 and M2 correspond to the $\text{Ga}^{3+}/\text{Al}^{3+}$ ions at the $18f$ (or $9b$) site and the Ge^{4+} ions at the $18f$ (or $9b$) site, respectively.

the contribution from low d_{bulk} is discussed in the next section. Note that the DSC analyses on $x = 0.5$ and 1 also showed no exothermic or endothermic peak up to 1300 K (Fig. S5 of the Supplementary Material).

Discussion

x dependence of $d_{\text{Li-O}}$ and $d_{\text{M-O}}$.—In the phenacite structure with $R\bar{3}$ space group, Li^+ ions are coordinated with four different O^{2-} ions, i.e., the O1, O2, O3, and O4 atoms. Moreover, $\text{Ga}^{3+}/\text{Al}^{3+}/\text{Ge}^{4+}$ ions at the Ga1/Al1/Ge1 ($18f$) site are coordinated with the O1, O2, and two O4 atoms, while $\text{Ga}^{3+}/\text{Al}^{3+}/\text{Ge}^{4+}$ ions at the Ga2/Al2/Ge2 ($18f$) site are coordinated with the O1, O2, and two O3 atoms. Such an environmental difference would alter the bond length between the Li^+ and O^{2-} ions in the LiO_4 tetrahedron ($d_{\text{Li-O}}$) and/or the bond length between the $\text{Ga}^{3+}/\text{Al}^{3+}/\text{Ge}^{4+}$ ($=\text{M}$) and O^{2-} ions ($d_{\text{M-O}}$) in the MO_4 tetrahedron.

Figure 7a shows the averaged $d_{\text{Li-O}}$ values for the LGAGO samples, which were calculated by d values of $d_{\text{Li-O1}}$, $d_{\text{Li-O2}}$, $d_{\text{Li-O3}}$, and $d_{\text{Li-O4}}$ for $R\bar{3}$ space group, or d values of $d_{\text{Li-O1}}$, $d_{\text{Li-O2}}$, $d_{\text{Li-O3}}$, $d_{\text{Li-O4}}$, $d_{\text{Li-O5}}$, $d_{\text{Li-O6}}$, $d_{\text{Li-O7}}$, and $d_{\text{Li-O8}}$ for $R3$ space group. The averaged $d_{\text{Li-O}}$ is almost constant ($\cong 1.92\text{ Å}$) except for the $x = 0.25$ sample. This value is comparable with the calculated $d_{\text{Li-O}}$ ($= 1.97\text{ Å}$) using $r_{\text{Li}^+} = 0.59\text{ Å}$ and $r_{\text{O}^{2-}} = 1.38\text{ Å}$ when C. N. = 4.²⁴ Conversely, as seen in Fig. 7b, $d_{\text{M-O}}$ splits into two different values ($d_{\text{M1-O}}$ and $d_{\text{M2-O}}$) when x increases from 0. That is, $d_{\text{M1-O}} = 1.801(5)\text{ Å}$ and $d_{\text{M2-O}} = 1.798(5)\text{ Å}$ at $x = 0$, but $d_{\text{M1-O}} = 1.759(6)\text{ Å}$ and $d_{\text{M2-O}} = 1.807(4)\text{ Å}$ at $x = 0.25$. Since $d_{\text{Al-O}} = 1.736(5)\text{ Å}$ and $d_{\text{Ge-O}} = 1.839(5)\text{ Å}$ at $x = 1$, M1 and M2 are attributed to the $\text{Ga}^{3+}/\text{Al}^{3+}$ ions at the $18f$ (or $9b$) site and the Ge^{4+} ions at the $18f$ (or $9b$) site, respectively.

According to previous reports on LiGaSiO_4 ³ and LiAlSiO_4 ,²⁶ $d_{\text{Li-O}} = 1.945\text{ Å}$ for the former and $d_{\text{Li-O}} = 1.982\text{ Å}$ for the latter. This indicates that $d_{\text{Li-O}}$ in phenacite compounds is not changed significantly by M ions at the $18f$ (or $9b$) site. The splitting into two different d values of $d_{\text{M1-O}}$ and $d_{\text{M2-O}}$ is likely to adjust with the decrease in a_h and c_h with x . It should be noted that some O–Li–O bond angles ($\phi_{\text{O-Li-O}}$) drastically changes with x . There are six different $\phi_{\text{O-Li-O}}$ s in the LiO_4 tetrahedron with $R\bar{3}$ space group, i.e., $\phi_{\text{O1-Li-O2}}$, $\phi_{\text{O1-Li-O3}}$, $\phi_{\text{O1-Li-O4}}$, $\phi_{\text{O2-Li-O3}}$, $\phi_{\text{O2-Li-O4}}$, and $\phi_{\text{O3-Li-O4}}$. When x increases to 0.25 from 0, $\phi_{\text{O1-Li-O2}}$ decreases to $100.3(1)^\circ$ from $114.0(2)^\circ$, whereas $\phi_{\text{O3-Li-O4}}$ increases to $130.6(4)^\circ$ from $113.2(7)^\circ$. These $\phi_{\text{O-Li-O}}$ values were determined by VESTA.¹⁹ Here, when $\phi_{\text{O-Li-O}} = 109.47^\circ$, the LiO_4 is the regular tetrahedron. Thus, the deformation in the LiO_4 tetrahedron also contributes

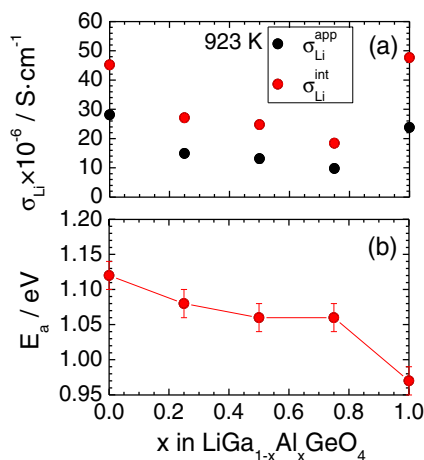


Figure 8. (a) Apparent Li-ion conductivity ($\sigma_{\text{Li}}^{\text{app}}$) and intrinsic Li-ion conductivity ($\sigma_{\text{Li}}^{\text{int}}$) and (b) activation energy (E_a) for Li-ion conduction as a function of x in $\text{LiGa}_{1-x}\text{Al}_x\text{GeO}_4$.

to the adjustment with the overall structural changes in the LGAGO lattice.

x dependence of $\sigma_{\text{Li}}^{\text{int}}$ and E_a .—Solid electrolytes, such as LLTO¹⁴ and $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$,¹⁵ have typically high $d_{\text{bulk}}/d_{\text{XRD}}$ values which are close to 1. On the contrary, $d_{\text{bulk}}/d_{\text{XRD}}$ for the present LGAGO samples and $\text{Li}[\text{Li}_{1/3}\text{Ti}_{5/3}]\text{O}_4$ (LTO) are lower than ~ 0.7 , due to the high sintering-resistance during the synthesis.¹⁷ Here, LTO has a spinel structure with $Fd\bar{3}m$ space group, in which Li^+ ions occupy both tetrahedral $8a$ and octahedral $16d$ sites.¹⁷ To compare with σ_{Li} values having different $d_{\text{bulk}}/d_{\text{XRD}}$ values, we recently clarified the effect of $d_{\text{bulk}}/d_{\text{XRD}}$ on the σ_{Li} of LTO.¹⁷ The apparent σ_{Li} ($\sigma_{\text{Li}}^{\text{app}}$), i.e., observed σ_{Li} is given by

$$\sigma_{\text{Li}}^{\text{int}} = \left(\frac{3\pi}{8\zeta} - \frac{3\pi}{8} + 1 \right) \sigma_{\text{Li}}^{\text{app}} \quad [2]$$

where ζ is $d_{\text{bulk}}/d_{\text{XRD}}$, and $\sigma_{\text{Li}}^{\text{int}}$ is intrinsic σ_{Li} .

By applying to the relationship between ζ and $\sigma_{\text{Li}}^{\text{app}}$, $\sigma_{\text{Li}}^{\text{int}}$ values for the LGAGO samples with $0 \leq x \leq 1$ were calculated. Figure 8a shows the x dependence of $\sigma_{\text{Li}}^{\text{int}}$ at 923 K as well as $\sigma_{\text{Li}}^{\text{app}}$ at the same T . As x increases from 0, $\sigma_{\text{Li}}^{\text{app}}$ decreases down to $1.84(1) \times 10^{-5} \text{ S}\cdot\text{cm}^{-1}$ at $x = 0.75$, and then rapidly increases to $4.77(1) \times 10^{-5} \text{ S}\cdot\text{cm}^{-1}$ at $x = 1$. $\sigma_{\text{Li}}^{\text{int}}$ is greater than $\sigma_{\text{Li}}^{\text{app}}$ in the whole x range, but the ratio of $\sigma_{\text{Li}}^{\text{int}}/\sigma_{\text{Li}}^{\text{app}}$ is different between the five LGAGO samples; $\sigma_{\text{Li}}^{\text{int}}$ for $x = 0$ is about 1.5 times larger than $\sigma_{\text{Li}}^{\text{app}}$, while $\sigma_{\text{Li}}^{\text{int}}$ for $x = 1$ is about 2 times larger than $\sigma_{\text{Li}}^{\text{app}}$. The $x = 1$ sample, thus, exhibits the maximum $\sigma_{\text{Li}}^{\text{int}}$ value ($= 4.77(1) \times 10^{-5} \text{ S}\cdot\text{cm}^{-1}$) among the five LGAGO samples.

As seen in Fig. 8b, the activation energy (E_a) for Li-ion conduction is lowest for the $x = 1$ composition. E_a is almost constant ($\cong 1.05 \text{ eV}$) up to $x = 0.75$, and then slightly decreases to $\sim 0.97 \text{ eV}$ at $x = 1$.

Possible conduction pathway in LGAGO.—Since all the Li^+ ions in LGAGO are tetrahedrally coordinated with O^{2-} ions, the Li^+ ions must firstly be removed from the LiO_4 tetrahedron to bring about the continuous Li-ion conduction. The bottleneck radius (r_{tet}) of the LiO_4 tetrahedron is calculated by

$$r_{\text{tet}} = \frac{l_{\text{tet}}}{\sqrt{3}} - r_{\text{O}^{2-}}, \quad [3]$$

where l_{tet} is the length of one side of the regular tetrahedron. Assuming that LiO_4 is the regular tetrahedron, r_{tet} is represented by

$$r_{\text{tet}} = \frac{2\sqrt{2}}{3} d_{\text{Li-O}} - r_{\text{O}^{2-}} \quad [4]$$

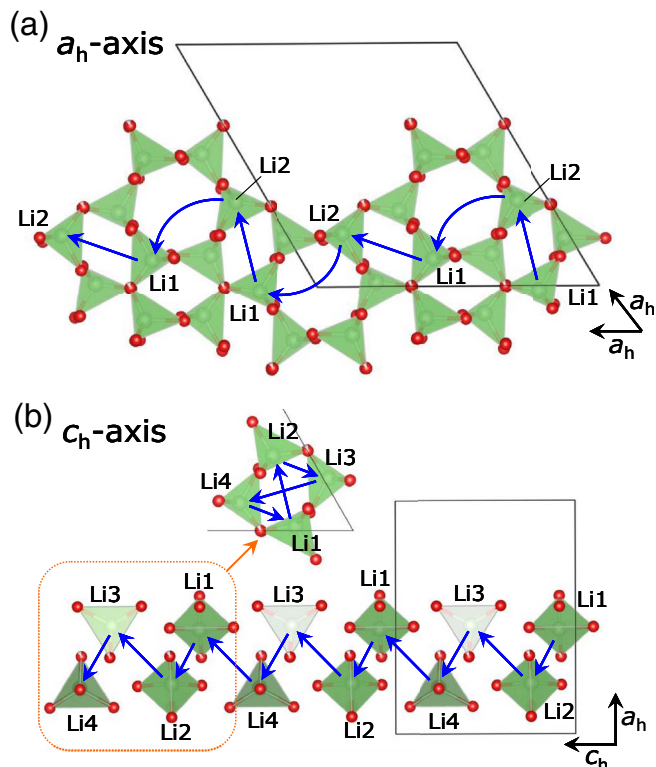


Figure 9. Possible conduction pathway for Li^+ ions in $\text{LiGa}_{1-x}\text{Al}_x\text{GeO}_4$ along the (a) a_h - and (b) c_h -axes. For conduction in (a), Li ion moves across between the Li1 to the Li2 positions. For conduction in (b), Li ion follows a zig zag pathway from the Li1 $\rightarrow$ Li2 $\rightarrow$ Li3 $\rightarrow$ Li4 positions. Inset of (b) indicates the schematics from the viewpoint of c_h -axis. Only the LiO_4 tetrahedra are shown for clarity.

r_{tet} at room- T is determined as 0.43(1), 0.48(1), 0.44(1), 0.45(1), and 0.43(1) Å for the $x = 0, 0.25, 0.5, 0.75$, and 1 samples, respectively, using the $d_{\text{Li-O}}$ values shown in Fig. 7a. These r_{tet} values are slightly larger than that of LTO ($\cong 0.39 \text{ Å}$),²⁷ but significantly lower than that of LLTO ($\cong 0.75 \text{ Å}$).¹⁴

Considering only the bond lengths between the Li atoms ($d_{\text{Li-Li}}$), $d_{\text{Li-Li}}$ for the 1st nearest neighboring (NN) Li ion ($d_{\text{Li-Li}}^{\text{1st}}$) is 4.00(2) Å with C. N. = 1. In addition, $d_{\text{Li-Li}}$ for the 2nd NN Li ion ($d_{\text{Li-Li}}^{\text{2nd}}$) is 4.33(3) Å with C. N. = 2, while that for the 3rd NN Li ion ($d_{\text{Li-Li}}^{\text{3rd}}$) is 4.91(2) Å with C. N. = 1. The $d_{\text{Li-Li}}$ values above are all for the $x = 0$ sample. However, for the $x = 1$ sample, $d_{\text{Li-Li}}^{\text{1st}} = 3.82(2) \text{ Å}$ with C. N. = 1, $d_{\text{Li-Li}}^{\text{2nd}} = 4.44(3) \text{ Å}$ with C. N. = 2, and $d_{\text{Li-Li}}^{\text{3rd}} = 4.82(2) \text{ Å}$ with C. N. = 2, in accordance with the decrease in a_h and c_h . From these Li coordinations, two conduction pathways of Li^+ ions are possible for LGAGO, as illustrated in Fig. 9. For the conduction along the a_h -axis (Fig. 9a), the Li ion moves in the Li1 $\rightarrow$ Li2 direction, followed by an opposite movement in the Li2 $\rightarrow$ Li1 direction. The distance from the Li1 to the Li2 positions ($d_{\text{Li1} \rightarrow \text{Li2}}$) is 4.00(2) Å for the $x = 0$ sample, which is equal to its $d_{\text{Li-Li}}^{\text{1st}}$. Although the direct distance from the Li2 to Li1 positions ($d_{\text{Li2} \rightarrow \text{Li1}}$) for the $x = 0$ sample is 5.04(2) Å, the Li ion must take a detour from the Li2 to the Li1 positions. This is because the movement from the Li2 to the Li1 positions is within six- LiO_4 rings, whereas that from the Li1 to Li2 positions is within four- LiO_4 rings.

For conduction along the c_h -axis shown in Fig. 9b, the Li ion follows a zig zag pathway through the Li1 $\rightarrow$ Li2 $\rightarrow$ Li3 $\rightarrow$ Li4 positions. For the $x = 0$ sample, $d_{\text{Li1} \rightarrow \text{Li2}}$, $d_{\text{Li2} \rightarrow \text{Li3}}$, $d_{\text{Li3} \rightarrow \text{Li4}}$, and $d_{\text{Li4} \rightarrow \text{Li1}}$ are 4.00(2), 4.33(3), 4.91(2), and 4.33(3) Å, respectively, which correspond to $d_{\text{Li-Li}}^{\text{1st}}$, $d_{\text{Li-Li}}^{\text{2nd}}$, $d_{\text{Li-Li}}^{\text{3rd}}$, and $d_{\text{Li-Li}}^{\text{2nd}}$, respectively. As seen in the inset of Fig. 9b, the movements from the Li1 $\rightarrow$ Li2 $\rightarrow$ Li3 $\rightarrow$ Li4

positions are all within four-LiO₄ rings. Therefore, the difference in the conduction pathway between the *a_h*- and *c_h*-axes depends on the movement from the Li2 position; Li2 → Li1 for the former, and Li2 → Li3 for the latter.

It should be noted that the carrier concentration of Li⁺ ions is not so different between the LGAGO samples. This is because, as described above, the number of NN Li ion and the amount of vacancy for the Li-ion conduction are almost similar between the LGAGO samples. This indicates that σ_{Li} for the LGAGO samples mainly depends on their structural environment such as bond lengths between Li ions.

Correlation between crystal structure and $\sigma_{\text{Li}}^{\text{int}}$ (E_a).—As shown in Fig. 8a, the $\sigma_{\text{Li}}^{\text{int}}$ values at $0.25 \leq x \leq 0.75$ are slightly lower than those at $x = 0$ and 1. There seems to be two opposite factors for determining $\sigma_{\text{Li}}^{\text{int}}$ for the LGAGO samples with $0.25 \leq x \leq 0.75$. One is a decrease in $d_{\text{Li-Li}}$ s, such as $d_{\text{Li1} \rightarrow \text{Li2}}$ and $d_{\text{Li2} \rightarrow \text{Li1}}$, and the other is an underdeveloped cation order between Ga³⁺/Al³⁺ and Ge⁴⁺ ions. The decrease in $d_{\text{Li-Li}}$ s, in other words, the decrease in *a_h*- and *c_h*-axis lengths unambiguously plays a favorable role on $\sigma_{\text{Li}}^{\text{int}}$. Conversely, at $0.25 \leq x \leq 0.75$, the Ga³⁺ and Al³⁺ ions occupy the same 9b or 18f site with the ratio of Ga³⁺/Al³⁺ = (1 - *x*)/*x* (see Tables I and S1). Furthermore, the FWHM of the 140 diffraction line for these compositions are slightly larger than those for $x = 0$ and 1. Thus, the underdeveloped cation order between Ga³⁺/Al³⁺ and Ge⁴⁺ ions produces a local structural distortion, resulting in the decrease in $\sigma_{\text{Li}}^{\text{int}}$. The $x = 1$ sample, which has the developed cation order between Al³⁺ and Ge⁴⁺ ions, exhibits the minimum E_a value, as seen in Fig. 8b. Although its origin is currently unclear, electrostatic repulsion between Li⁺ and Al³⁺ (or Ge⁴⁺) ions would play a significant role in determining E_a for the LGAGO samples.

Even for the $x = 1$ sample, its $\sigma_{\text{Li}}^{\text{int}}$ (E_a) is lower (higher) than that for LTO, i.e., $\sigma_{\text{Li}}^{\text{int}} = 7.4 \times 10^{-3} \text{ S}\cdot\text{cm}^{-1}$ at 923 K and $r_{\text{tet}} \cong 0.70 \text{ eV}$ in the *T* range between 773 and 923 K.²⁶ Note that r_{tet} for the $x = 1$ sample ($= 0.43(1) \text{ \AA}$) is almost comparable with that for LTO ($\cong 0.39 \text{ \AA}$), as described in Possible conduction pathway in LGAGO section. Here, LTO exhibits a zig zag conduction pathway through the $8a \rightarrow 16c \rightarrow 8a$ direction, where 16c is the vacant octahedral site in the spinel structure.²⁷ The Li⁺ ions at the 8a site are tetrahedrally coordinated with the vacant 16c site with the bond length of $d_{8a-16c} = 1.81(1) \text{ \AA}$. Thus, the number of Li⁺ ions in the NNs, i.e., the carrier density of Li⁺ ions, and their $d_{\text{Li-Li}}$ s are crucial for determining $\sigma_{\text{Li}}^{\text{int}}$ (E_a) for the phenacite and spinel compounds. Since LiAlGeO₄ is reported to transform into the spinel structure under high pressures above 7 GPa,⁴ EIS measurements for that compound would provide further information of such correlation.

It is interesting to compare with the results for Li_{2.5}Ga_{0.5}GeO₄¹² and Li_{4-3*y*}Al_{*y*}GeO₄,¹³ because Li_{4-3*y*}Ga_{*y*}GeO₄ with $y = 0.5$ (or Li_{4-3*y*}Al_{*y*}GeO₄) is regarded as a solid-solution compound between LiGaGeO₄ (or LiAlGeO₄) and Li₄GeO₄. The σ_{Li} value of Li₄GeO₄ is $8.7 \times 10^{-5} \text{ S}\cdot\text{cm}^{-1}$ at 673 K,²⁸ but the substitution of Li⁺ ions for Ga³⁺ (or Al³⁺) ions significantly increases its σ_{Li} up to $\sim 1.0 \times 10^{-1} \text{ S}\cdot\text{cm}^{-1}$ at the same *T*, due to the partial occupation of the Li⁺ ions at the Ga³⁺ (or Al³⁺) ions' site.^{12,13} In the LGAGO lattice with $0 \leq x \leq 1$, Li⁺ ions do not occupy the Ga³⁺, Al³⁺, or Ge⁴⁺ ions' site. Therefore, the substitution of these ions for Li⁺ ions would be useful for increasing (decreasing) the σ_{Li} (E_a) values for the LGAGO compounds.

Conclusions

The LiGa_{1-*x*}Al_{*x*}GeO₄ (LGAGO) samples with $x = 0, 0.25, 0.5, 0.75, 1$ were prepared by a solid-state reaction technique and their crystal structures were investigated by powder X-ray diffraction (XRD) measurements. The Rietveld refinement on the $x = 0$ sample showed that the crystal lattice is identified as a phenacite structure with *R* $\bar{3}$ space group, in which Li⁺ ions occupy tetrahedral 18f site, Ga³⁺ and Ge⁴⁺ ions occupy two nonequivalent tetrahedral 18f sites, and O²⁻ ions occupy four nonequivalent 18f sites. Here, Ga³⁺ and Ge⁴⁺ ions

are randomly distributed at the two 18f sites. As *x* increased from 0, i.e., as the amount of Al³⁺ ions increased, the lattice parameters in the hexagonal setting (*a_h* and *c_h*) decreased almost linearly, in accordance with the ionic radii (*r*) of the Ga³⁺ ($r_{\text{Ga}^{3+}} = 0.47 \text{ \AA}$) and Al³⁺ ($r_{\text{Al}^{3+}} = 0.39 \text{ \AA}$) ions. Furthermore, the crystal structure for the $x \geq 0.5$ sample was assigned as an ordered phenacite structure with *R*3 space group, in which Ga³⁺/Al³⁺ and Ge⁴⁺ ions arrange along the *c_h*-axis. Due to this cation order, the bond length between Ga³⁺/Al³⁺/Ge⁴⁺ (= *M*) and O²⁻ ions (*d_{M-O}*) clearly presented two different values of *d_{M1-O}* ($\cong 1.75 \text{ \AA}$) and *d_{M2-O}* ($\cong 1.85 \text{ \AA}$) at $x \geq 0.5$.

We also examined Li⁺-ion conductivity (σ_{Li}) for the LGAGO samples by electrochemical impedance measurements. σ_{Li} for the $x = 0$ sample is $1.15(1) \times 10^{-7} \text{ S}\cdot\text{cm}^{-1}$ at 673 K and $2.81(1) \times 10^{-5} \text{ S}\cdot\text{cm}^{-1}$ at 973 K. Since the ratio of the bulk density (*d_{bulk}*) to the XRD density (*d_{XRD}*) is limited to be 0.66(1) for the $x = 0$ sample, we calculated intrinsic σ_{Li} ($\sigma_{\text{Li}}^{\text{int}}$) using the relation between *d_{bulk}*/*d_{XRD}* and observed σ_{Li} . $\sigma_{\text{Li}}^{\text{int}}$ for the $x = 0$ sample was increased to $4.51(1) \times 10^{-5} \text{ S}\cdot\text{cm}^{-1}$ at 973 K. As *x* increases from 0, $\sigma_{\text{Li}}^{\text{int}}$ at 973 K slightly drops to $\sim 2.5 \times 10^{-5} \text{ S}\cdot\text{cm}^{-1}$ at $0.25 \leq x \leq 0.75$, and then increases to $4.77(1) \times 10^{-5} \text{ S}\cdot\text{cm}^{-1}$ at $x = 1$.

Although the cation order between Al³⁺ and Ge⁴⁺ ions increases (decreases) $\sigma_{\text{Li}}^{\text{int}}$ (E_a) for the LGAGO samples, their σ_{Li} values are still much lower than those for related compounds γ -Li_{4-3*x*}Al_{*x*}GeO₄ and Li[Li_{1/3}Ti_{5/3}]O₄. The structural consideration clarified that the large distance between the Li ions in LGAGO, the number of Li⁺ ions in the nearest neighbors decreases (increases) $\sigma_{\text{Li}}^{\text{int}}$ (E_a). The advantage of LGAGO except for the low-temperature synthesis is currently unknown. Studies on voltage window and chemical or electrochemical stability are underway in our laboratory.

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